MILITARY SPECIFICATION

INDICATOR, TEMPERATURE, THERMOCOUPLE, EHU-43/A

This specification is approved for use by the Department of the Air Force and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements for single-pointer, self-balancing, potentiometer-type temperature indicator type EHU-43/A dc white lighted (12 pin receptacle).

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein:

SPECIFICATIONS

MILITARY

MIL-I-83262	Indicator, Temperature, Thermocouple, Hermetically Sealed, General Specification for
MIL-C-83723	Connector, Electrical, (Circular, Environment Resisting), Receptacles and Plugs, General Specification for

STANDARDS

FEDERAL

FED-STD-595	Colors

MILITARY

MIL-STD-129	Marking for Shipment and Storage
MS33585	Pointers, Dial, Standard Design of Aircraft Instrument
MS33639	Cases, Instrument, Clamp-Mounted, Aircraft

FSC 6685

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: ASD/ENESS, Wright-Patterson AFB, OH 45433 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

Source: https://assist.dla.mil -- Downloaded: 202610-23T07:44Z
Check the source to verify that this is the current version before use.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 General specification. Unless otherwise specified herein, the requirements of MIL-I-83262 shall apply. In case of conflict between MIL-I-83262 and this specification, this specification shall govern.

3.2 Certification. The contractor shall certify that the indicators furnished under this specification have passed the qualification tests in accordance with MIL-I-83262.

3.2.1 Power consumption. The indicator steady-state power consumption shall not exceed 7Va exclusive of the lighting circuit.

3.2.2 Reliability. The indicator shall have a minimum acceptable mean-time-between-failures (MTBF) of at least 2,000 hours at a 90 percent confidence level.

3.3 Design and construction. The indicator shall incorporate a self-balancing potentiometer-type circuit for temperature measurement. The indicators shall include a dc torque motor. The indicator thermocouple input impedance shall be sufficiently high so that two indicators may be operated simultaneously, each within the performance requirements of this specification, when wired in parallel from a single chromel-alumel thermocouple constructed of 0.050 inch diameter wire or smaller.

3.3.1 Power supply. The power supply shall be in accordance with MIL-I-83262 for the dc powered indicators.

3.4 Performance. The indicator shall meet the following requirements:

a. Scale error not exceeding the tolerances specified in table I

b. Friction error not exceeding 2°C over a temperature range of 710° to 990°C

c. Position error not exceeding 1°C over a temperature range of 710° to 990°C

d. Case leakage not exceeding 0.1 micron cubic foot per hour

e. Warmup and pointer travel from 100° to 1,000°C within 3 seconds at room temperature and 6 seconds at -55° and +100°C

f. Instantaneous indication change from 500° to 850°C in not more than 2 seconds at room temperature and 2.5 seconds at -55° to +100°C

Source: https://assist.dla.mil -- Downloaded: 202610-23T07:44Z
Check the source to verify that this is the current version before use.

g. When the signal is increased or decreased 0.08 mv, a change of 2° ±1°C indication shall be noted at points between 710° and 990°C

h. Withstand temperatures ranging from -65° to +100°C

i. Withstand vibration (see 4.1.4.8, 4.1.4.8.1, and 4.1.4.8.2)

j. Withstand an alternate input of -3 and +52 mv for a period of 30 minutes at each condition for eight cycles with the ambient temperature at 50°C

k. Indication change not exceeding 2°C when a lead resistance change from 5 to 200 ohms is made at specified test points between 710° and 990°C

l. Indication steady-state change not exceeding 2°C when either the chromel or alumel conductor is grounded separately at any point to the ground side (low potential side) of the 28V power supply when the indicator is operated in the range of 710° to 790°C

m. Withstand pressures ranging from 30 inches Hg down to 0.309 inch Hg, (approximately sea level to 100,000 feet)

n. Withstand voltage variations from 16V to 30V dc with a ripple about the mean voltage, and voltage surges up to 80V dc

o. Withstand eight cycles of alternate immersion in tap water at 85° ±2°C and 5° ±2°C

p. Withstand a mechanical shock of 10 g for 11 ±1 milliseconds with no damage or malfunction of any component of the indicator

q. Exposure to salt sea atmosphere for a period of 48 hours.

TABLE I. Test tolerances.

Test Points °C	Scale Error at Room Temperature ±°C	Scale Error at Low Temperature ±°C	Scale Error at High Temperature ±°C	Scale Error During Reliability Test ±°C	Friction Error °C
100	15	20	20	20	---
200	15	20	20	20	---
400	15	20	20	20	---
600	15	20	20	20	---
710	5	10	10	10	2
800	5	10	10	10	2
900	5	10	10	10	2
990	5	10	10	10	2
1,000	15	20	20	20	---
1,100	15	20	20	20	---
1,200	15	20	20	20	---

Source: https://assist.dla.mil -- Downloaded: 202610-23T07:44Z
Check the source to verify that this is the current version before use.

3.5 Case. The indicator case shall be hermetically sealed in accordance with MS33639 for the 2-inch diameter size. The case length shall be 7 ±0.25 inches exclusive of the electrical receptacle. The bezel and that part of the case visible from the cockpit side of the instrument panel shall be finished in lusterless black, Color No. 37038 of FED-STD-595. The remainder of the case may be finished in black and shall be adequately protected to satisfactorily withstand the environmental test conditions specified herein. The hermetic seal shall not be accomplished by use of materials that are affected by the action of any atmosphere to which the case may be subjected. The front edge of the case shall not extend beyond the outer surface of the cover glass by more than .03125 inch. The case shall be so designed that the internal mechanism may be removed from the case, replaced, and the case resealed without the use of special tools unless they are approved by the procuring activity.

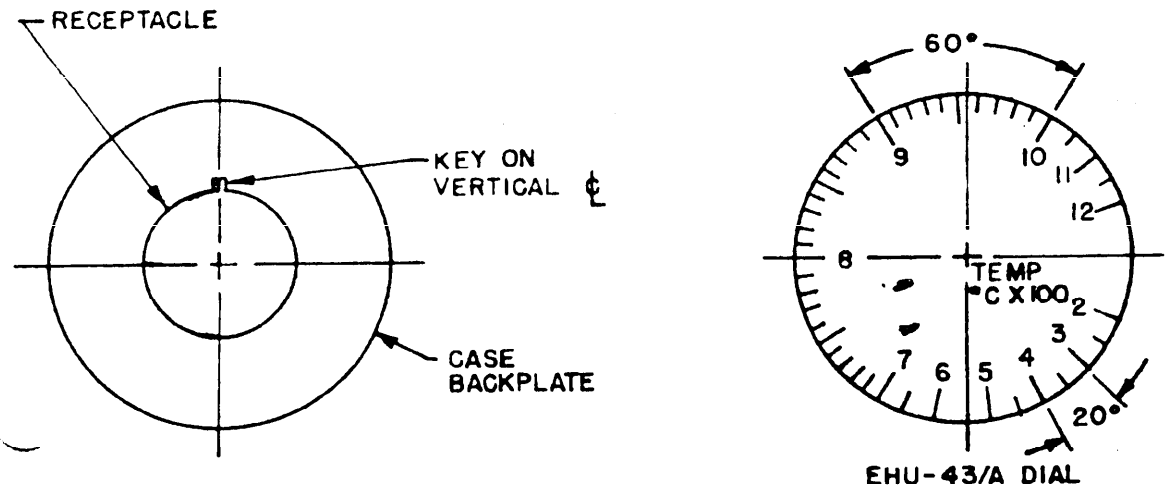
3.6 Dial. The indicator dial shall be as shown on figure 1 and as otherwise specified herein. A minimum diameter of 1.75 inches shall be maintained across the outer ends of the scale graduations. The total scale arc shall be 340° ±2°. Within the contracted and expanded portions of the scale, the scale shall be within 0.5 angular degree of precise linear spacing between adjacent scale graduations. The background of the dial shall be durable lusterless black consistent with integral lighting requirements.

3.7 Pointer. The pointer shall be in accordance with MS33585-8, except the tip of the pointer shall extend to the inner ends of the major scale graduations ±0.016 inch. The length of the lighted portion of the pointer shall be not less than 0.060 inch.

3.8 Marking dimensions and colors. The dial and pointer colors shall be in accordance with FED-STD-595. The dimensions of the numerals and letters shall be as follows:

	Height or Length Inch ± 0.005	Width of Line or Graduation Inch ± 0.005	Color
Numerals	0.156	0.031	37875
50°C graduations (700° to 1,000°C)	0.156	0.031	37875
100°C graduations	0.156	0.031	37875
50°C graduations (except 700° to 1,000°C)	0.125	0.020	37875
Lettering TEMP °C x 100	0.094	0.020	37875
10°C graduations	0.125	0.020	37875
Background of dial	---	---	37038
Pointer	---	---	37875

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T07:44Z
Check the source to verify that this is the current version before use.



NOTE:
DIMENSIONS IN INCHES. UNLESS OTHERWISE SPECIFIED, TOLERANCES: ANGLES ± 0° 15'.

FIGURE 1. Dial and case backplate.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T07:44Z
Check the source to verify that this is the current version before use.

3.9 Electrical receptacle. The indicator shall be provided with a receptacle in accordance with MIL-C-83723, series 3 with bayonet coupling of type M83723-80H12127 except that Pins 11 and 12 shall be made of alumel and chromel materials respectively. The mating receptacle shall be a type M83723-75R12127.

3.9.1 Indicator wiring. The following connections shall be used for the indicator:

Pin	External Connection
7	28V dc power
6	28V dc ground
11	Alumel lead wire
12	Chromel lead wire
5	Case ground
8	5V internal lighting
9	5V lighting ground

3.9.2 Grounding of case. The case shall be grounded by connecting a wire from the terminal of Pin 5 of the receptacle to a lug on the inner surface of the backplate of the indicator.

3.10 Excluded components. The design and fabrication of the indicator shall not include the following components:

a. Electromechanical choppers

b. Gears for motor movement transfer purposes unless specifically approved

c. Plastic packaged integrated circuits.

3.11 Weight. The weight of the indicator shall not exceed 1.25 pounds.

4. QUALITY ASSURANCE PROVISIONS

4.1 Sampling, inspection and test

4.1.1 Sampling. Sampling shall be in accordance with MIL-I-83262 except vibration testing shall be conducted as follows:

Functional Vibration - Conducted as a Sampling Plan A test.
Gunfire Vibration - Conducted as a Sampling Plan B test after the cold junction compensator lag tests and before the extreme low temperature test.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T07:44Z
Check the source to verify that this is the current version before use.

4.1.2 Qualification inspection. Inspection for qualification shall be in accordance with MIL-I-83262 and as specified herein. Unless otherwise specified herein, the contractor shall perform the inspection and submit certification that the indicators conform to the requirements of this specification. A copy of the qualification test results shall be furnished to the Government.

4.1.2.1 Certification. Certification is authorized for this item under the following conditions:

a. Contractor's responsibilities

(1) The contractor will be responsible for performance of all inspections and tests necessary to insure that the item submitted for Government acceptance is in accordance with all the requirements of the drawings and specifications listed in the contract.

(2) Prior to shipment of the item, the contractor shall submit to the Government a certification of conformance based on either the contractor's inspection and tests, or those conducted by another qualified laboratory or inspection organization.

(3) The certification shall delineate all drawing and specification requirements being attested to, and shall be signed by a responsible officer of the contractor with authority to bind his principle.

(4) The certification shall contain the following statement: "I certify that the item described in this document has been inspected and tested and meets the following requirements. (Delineation of applicable documents.)"

b. Government responsibilities. Item shall not be accepted until it has been properly identified with the certification. Government acceptance shall be made on the basis of contractor's certification of conformance; however, the Government reserves the right to independently inspect items for conformance to all contractual drawings and specification requirements before acceptance.

4.1.3 Acceptance inspection. Acceptance inspection shall be in accordance with MIL-I-83262 except as otherwise specified herein.

4.1.4 Test methods. Unless otherwise specified herein, all test methods and procedures shall be in accordance with MIL-I-83262.

4.1.4.1 Power supply. Unless otherwise specified, the indicator shall be tested with 28V dc power with 4.5 ±0.5V ac or dc applied to the lighting circuit.

Source: https://assist.dla.mil -- Downloaded: 202610-23T07:44Z
Check the source to verify that this is the current version before use.

4.1.4.2 Scale error and friction. The friction error test shall be in accordance with MIL-I-83262 and shall be inducted at 710°, 800°, 900°, and 990°C.

4.1.4.3 Position error. This test shall be conducted at the 710°, 800° and 990°C test points.

4.1.4.4 Response time. Millivoltage signals and power shall be applied to provide a stable indication at the low end of the expanded scale. The millivoltage shall then be increased to produce an instantaneous change equivalent to 350°C. The time required for the indicator to fully respond to the 350°C change shall be noted and shall not exceed 2 seconds. The test shall be repeated going from the high end of the expanded scale to the low end of the expanded scale. The same tolerance shall apply.

4.1.4.5 Sensitivity. The procedure shall be repeated at the 850° and 950°C test points.

4.1.4.6 Thermocouple lead resistance effect. This test shall be conducted at the 710°, 800°, and 900°C test points.

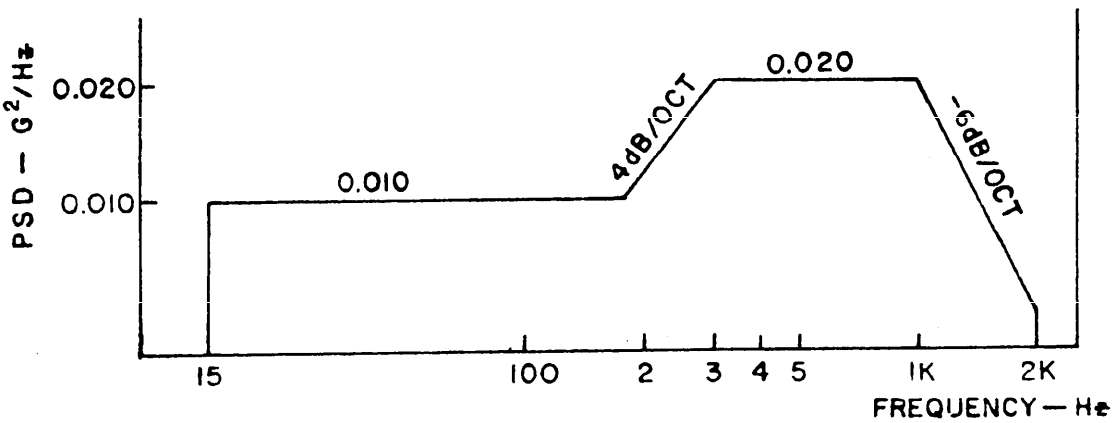
4.1.4.7 Grounding of thermocouple extension wires. Scale error shall be determined at the 710° and 890°C test points and the response time shall be determined in the 600° and 950°C range on increasing temperature.

4.1.4.8 Vibration. The indicator shall be subjected to the following tests.

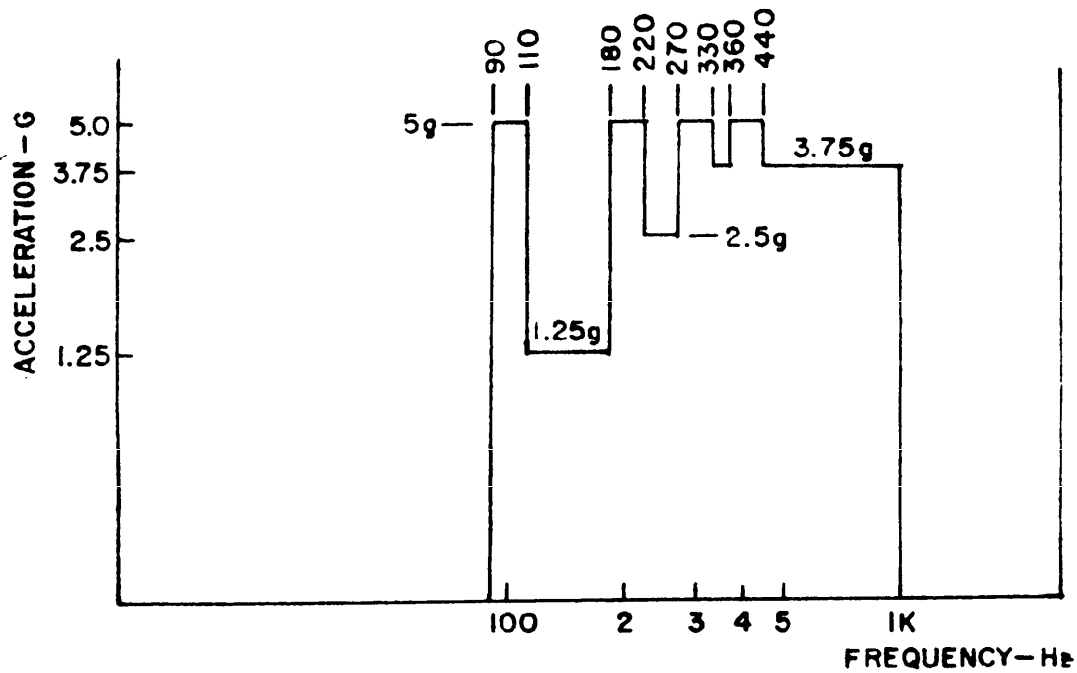
4.1.4.8.1 Functional vibration. The indicator shall be subjected to a random vibration test in accordance with MIL-STD-810, Method 514.2, equipment category b.2 except only a functional test level shall be performed and the endurance level of Procedure IA shall be omitted. The functional acceleration power spectral density shall be as shown on figure 2, Functional Vibration. The vibration requirements of MIL-I-83262 shall apply.

4.1.4.8.2 Gunfire vibration. The indicator shall be subjected to a vibration test in accordance with MIL-STD-810, Method 514.2, equipment category b.1 except the vibration test curve shall be as shown on figure 2, Gunfire Vibration. During the resonance dwell portion of the test the acceleration levels at the resonant dwell frequencies shall be as shown on figure 2, Gunfire Vibration. In addition to the dwell points found by MIL-STD-810 methods, the indicator shall be subjected to dwells at four additional dwell points. These points are 100 ±10 Hz, 200 ±20 Hz, 300 ±30 Hz, 400 ±40 Hz. At each of these frequencies the indicator shall be subjected to 10g acceleration for 5 minutes along each of the three mutually perpendicular indicator axes. The vibration requirements of MIL-I-83262 shall apply except pointer oscillation of ±10°C shall be acceptable.

Source: https://assist.dla.mil -- Downloaded: 202640-23T07:44Z
Check the source to verify that this is the currentversion before use.



FUNCTIONAL VIBRATION



GUNFIRE VIBRATION

FIGURE 2. Vibration tests.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T07:44Z
Check the source to verify that this is the current version before use.

5. PACKAGING

5.1 Preservation-packaging. Preservation-packaging shall be in accordance with MIL-I-83262.

5.2 Marking. In addition to any special marking required by the contract or order, interior and exterior containers shall be marked in accordance with MIL-STD-129.

5.2.1 Nomenclature. The nomenclature shall be as follows:

INDICATOR, TEMPERATURE
THERMOCOUPLE, SELF-BALANCING
POTENTIOMETER EHU-43/A

6. NOTES

6.1 All notes specified in MIL-I-83262 are applicable to this specification.

Custodian:
Air Force - 11

Reviewer activity:
Air Force - 99

Preparing activity:
Air Force - 11

Project No. 6685-F502

Source: https://assist.dla.mil -- Downloaded: 202610-23T07:44Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS: This form is provided to solicit beneficial comments which may improve this document and enhance its use. DoD contractors, government activities, manufacturers, vendors, or other prospective users of the document are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity. A response will be provided to the submitter, when name and address is provided, within 30 days indicating that the 1426 was received and when any appropriate action on it will be completed.

NOTE: This form shall not be used to submit requests for waivers, deviations or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

DOCUMENT IDENTIFIER *(Number)* AND TITLE

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

☐ VENDOR ☐ USER ☐ MANUFACTURER

1. ☐ HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE? ☐ IS ANY PART OF IT TOO RIGID, RESTRICTIVE, LOOSE OR AMBIGUOUS? PLEASE EXPLAIN BELOW.

A. GIVE PARAGRAPH NUMBER AND WORDING

B. RECOMMENDED WORDING CHANGE

C. REASON FOR RECOMMENDED CHANGE(S)

2. REMARKS

SUBMITTED BY *(Printed or typed name and address — Optional)*	TELEPHONE NO.
	DATE

DD FORM 1426 1 OCT 76 PREVIOUS EDITION WILL BE USED.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T07:44Z
Check the source to verify that this is the current version before use.

FOLD

ASD/ENESS
Wright-Patterson AFB, O 45433

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

POSTAGE AND FEES PAID
DEPARTMENT OF AIR FORCE
DOD-318



ASD/ENESS
Wright-Patterson AFB, Ohio 45433

FOLD

Source: https://assist.dla.mil -- Downloaded: 202640-23T07:44Z
Check the source to verify that this is the currentversion before use.